

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO3: *Examine cryptographic hash algorithms, digital signature schemes, and assess Kerberos and X.509 authentication frameworks for ensuring integrity, authentication, and non-repudiation.* CO (BL4 , BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 5

TOTAL MARKS: 25

EXPERIMENTS

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Experiments

1. Implement and compare MD5 and Secure Hash Algorithm (SHA-256) in Python by hashing multiple input messages. Analyze their collision resistance and performance differences.
2. Develop a Python program to implement a Digital Signature scheme using RSA. Demonstrate message signing and signature verification, and analyze its effectiveness in ensuring integrity and authentication.
3. Write a Python program to simulate HMAC (Hash-based Message Authentication Code) using a chosen hash function. Compare its security properties with a simple hash-based integrity check.
4. Implement a simplified Kerberos authentication mechanism in Python to simulate ticket generation, distribution, and validation between a client and server. Analyze its effectiveness in preventing replay attacks.
5. Develop a Python-based implementation of either the ElGamal or Schnorr digital signature scheme, and evaluate its security features compared to standard digital signature approaches.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation